

AS MATHEMATICS 9709 - PURE 1

Formula and method guide

The complete P1 memory book

YOUR AIM TODAY

Use one compact guide to select methods across all eight syllabus sections.

YOUR RULE

Do not memorise passively. Cover the right side, say the method aloud, then reproduce it from memory.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

Algebra, functions and coordinates

Foundations that appear throughout the paper.

Skill	Formula / method	Check
Complete square	$x^2+bx+c=(x+b/2)^2+c-(b/2)^2$	Expand back
Quadratic roots	$x=(-b\pm\sqrt{b^2-4ac})/(2a)$	Both roots
Discriminant	$D=b^2-4ac$: $D>0$ two, $D=0$ repeated, $D<0$ none	Use inequality
Composite	$fg(x)=f(g(x))$; do the inside function first	Order matters
Inverse	Write $y=f(x)$, swap x and y , rearrange	Check $f^{-1}(f(x))=x$
Gradient	$m=(y_2-y_1)/(x_2-x_1)$	Keep point order
Line	$y-y_1=m(x-x_1)$	Substitute point
Distance	$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$	Positive
Midpoint	$((x_1+x_2)/2, (y_1+y_2)/2)$	Between points
Perpendicular	$m_1*m_2=-1$	Negative reciprocal

CIRCLE BRIDGE

$(x-a)^2+(y-b)^2=r^2$ has centre (a,b) . A tangent is perpendicular to the radius at the contact point.

Circular measure and trigonometry

Angles must be in the correct unit.

Skill	Formula / method
Conversion	$180 \text{ degrees} = \pi \text{ radians}$; $\text{degrees} \times \pi/180$; $\text{radians} \times 180/\pi$
Arc	$s = r \theta$, with θ in radians
Sector	$A = (1/2)r^2 \theta$
Segment	$A = (1/2)r^2(\theta - \sin \theta)$
Identities	$\sin^2 x + \cos^2 x = 1$; $\tan x = \sin x / \cos x$
Exact values	Know $\sin$, $\cos$ and $\tan$ at 30, 45 and 60 degrees plus related angles
Graphs	State amplitude, period, range and transformations
Equation	Isolate trig function; reference angle; quadrants; interval check

GRAPH PERIODS

$\sin(kx)$ and $\cos(kx)$ have period $2\pi/k$. $\tan(kx)$ has period π/k .

Quadrant	sin	cos	tan
I	+	+	+
II	+	-	-
III	-	-	+
IV	-	+	-

Series

Choose AP, GP or binomial from the structure.

Model	Term	Sum / coefficient
AP	$u_n = a + (n-1)d$	$S_n = n/2[2a + (n-1)d] = n/2(a+l)$
GP	$u_n = ar^{(n-1)}$	$S_n = a(1-r^n)/(1-r)$
Infinite GP	Requires $ r < 1$	$S_{\infty} = a/(1-r)$
Binomial	Term $r+1$ uses $C(n,r)a^{(n-r)}b^r$	Track number, sign and x-power

SELECTION

Subtract consecutive terms to test an AP. Divide consecutive terms to test a GP. In a binomial expansion, r begins at zero.

Binomial checklist

- Write the general term before hunting a coefficient.
- For $(a-b)^n$, the sign depends on the power of $-b$.
- For a constant term involving x and $1/x$, set the total power of x equal to zero.

Differentiation and integration

Calculus connects gradient, change and area.

Skill	Method
Differentiate	$d/dx(x^n) = nx^{(n-1)}$; work term by term
Tangent	Find point; evaluate dy/dx ; use $y-y_1=m(x-x_1)$
Normal	Use negative reciprocal of tangent gradient
Stationary	Solve $dy/dx=0$; find y ; classify by sign or second derivative
Integrate	$\int x^n dx = x^{(n+1)}/(n+1) + C$, $n \neq -1$
Find C	Integrate first, then substitute the given point
Definite	Evaluate $F(\text{upper}) - F(\text{lower})$
Area	Sketch; split at crossings; area must be positive
Volume	About x-axis: $V = \pi \int y^2 dx$

CALCULUS CHAIN

Differentiate -> equation of gradient -> solve. Integrate -> primitive -> limits or constant -> interpret.

The 110-minute routine

Paper 1 contains 75 marks.

Time	Action
0-5 min	Scan; begin with accessible questions
5-95 min	Work at about 1.3 minutes per mark
95-105 min	Return to incomplete parts
105-110 min	Check signs, intervals, exact form and omitted parts

WHEN STUCK

Write the relevant identity, formula or derivative. Substitute known values. Leave space and move on. A visible method can earn marks.

Final check	Done
Both quadratic roots	
Trig solutions inside interval	
Radians used for arc/sector	
+C on indefinite integrals	
Exact form preserved	
Every question attempted	